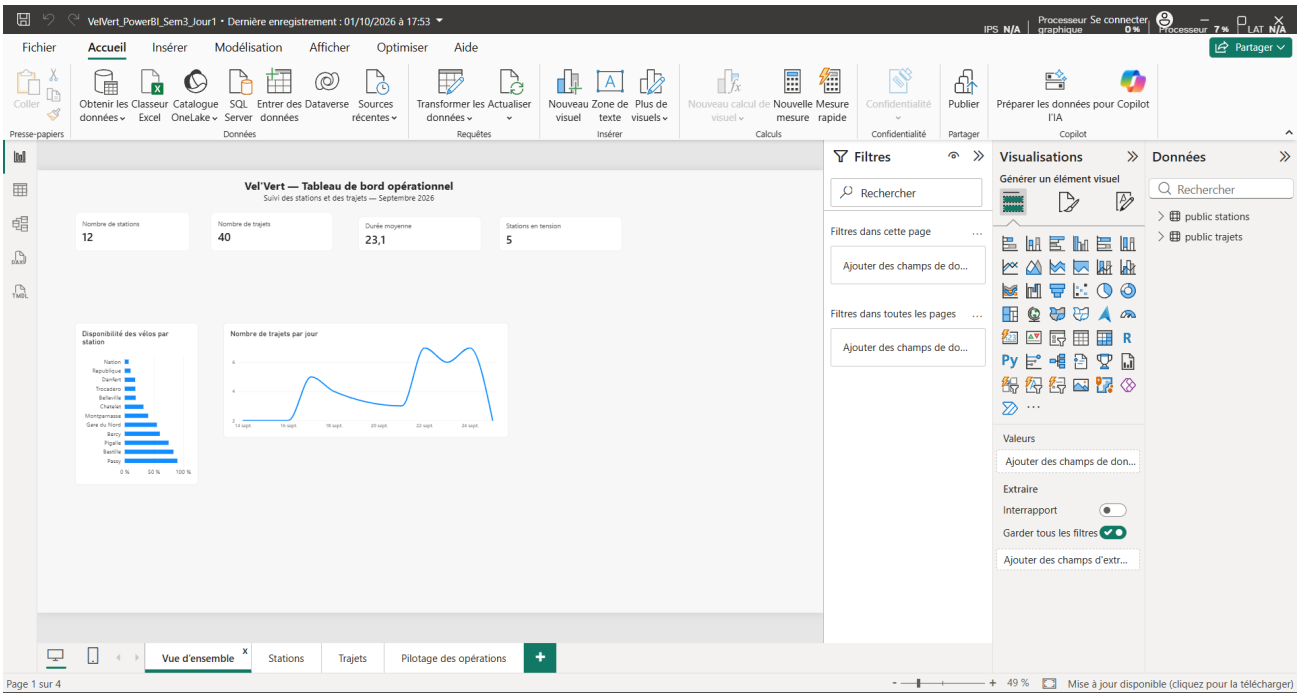
VEL'VERT

Operational Analysis of an Electric Bike-Sharing Service

SQL analysis	Dashboard	Business objective
PostgreSQL · DBeaver	Power BI · 4 report pages	Prioritise station replenishment



A practical data-analysis exercise translating operational questions into SQL queries and a Power BI report.

Training project — Simplon Data Analyst Programme, 2026.

1. Executive summary

Vel'Vert's operations team needs a quick way to identify stations that may require intervention and to review trip patterns. This project uses a small exercise dataset containing 12 stations and 40 trips. SQL queries filter, sort and compare station and trip attributes; the results are presented in a four-page Power BI dashboard.

12	40	5 / 12	23.1 min
Stations	Trips	Stations under pressure	Average trip duration

Main message

Five stations fall below the chosen threshold of 20% bike availability. The operational recommendation is to begin the next replenishment round with these stations, starting with the lowest availability rate.

Scope and method

The analysis follows the brief: inspect the data dictionary; write and execute commented SQL queries; use filters, sorting, limits and combined conditions; define an explicit criterion for the business question; and communicate the findings in plain language.

Tools

PostgreSQL, SQL, Python (collection script), DBeaver, Power BI and Git for local version control.

2. SQL analysis and operational findings

Data model

Table	Role in the analysis
stations	Station name, available bikes, total capacity, neighbourhood and installation date.
trajets	Trip identifier, departure station, trip duration and trip date.

Query patterns practised

• Selecting specific columns or all columns with **SELECT**.

• Filtering by values and categories with **WHERE**.

• Sorting with **ORDER BY** and limiting ranked results with **LIMIT**.

• Comparing available bikes with total capacity to identify availability below 20% or occupancy above 80%.

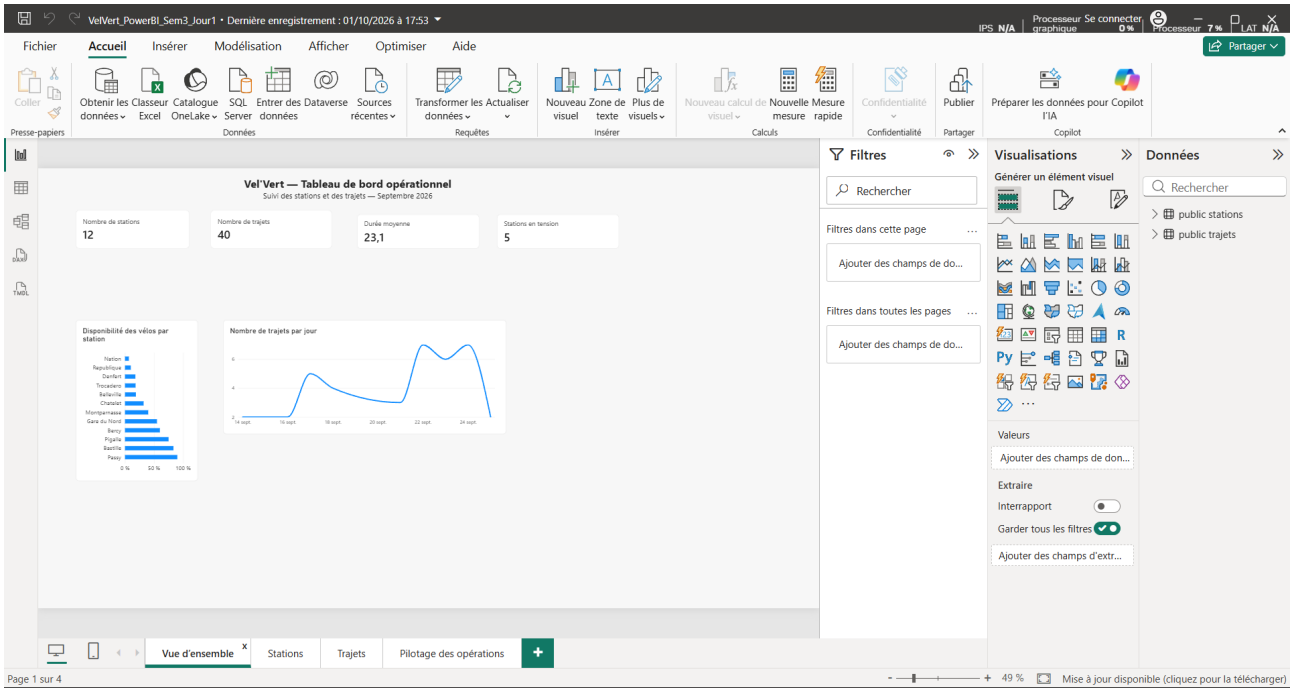
• Combining trip-duration and date conditions with **AND**, **OR** and range filters.

• Explaining the business criterion chosen for the final replenishment-priority query.

Key findings

Finding	What it means for operations
5 of 12 stations are below 20% availability.	These are the first candidates for replenishment; rank them by the lowest availability rate.
Overall availability rate shown: 41.02%.	This is a snapshot across the stations in the exercise dataset, not a forecast of future demand.
Average trip duration: approximately 23 minutes.	A useful descriptive baseline for reviewing trip-duration patterns.
Maximum trip duration shown: 66 minutes.	Long trips can be flagged for review, but duration alone does not prove an error.

3. Dashboard — Overview

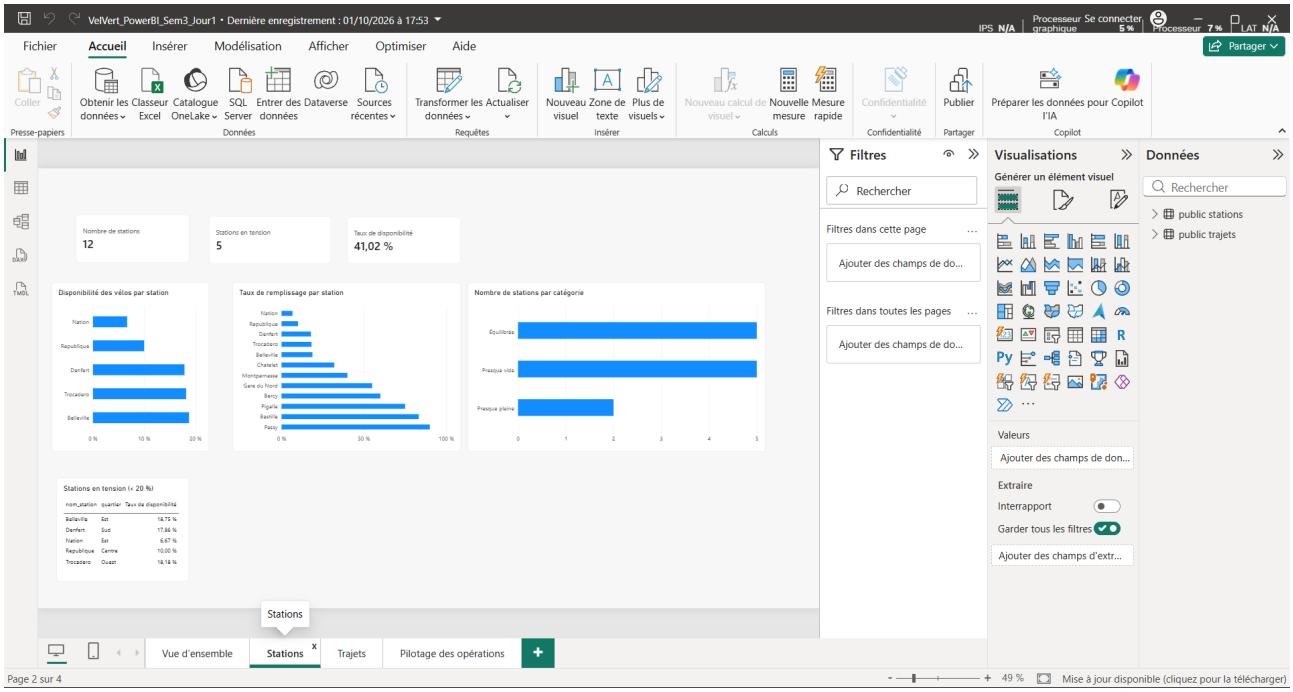


The overview page presents the number of stations and trips, average trip duration, stations under pressure, bike availability by station and daily trip counts.

Interpretation

The dashboard makes the core indicators visible at a glance. The station ranking is useful for prioritising action, while the daily trip trend provides context for the observed activity.

4. Dashboard — Stations

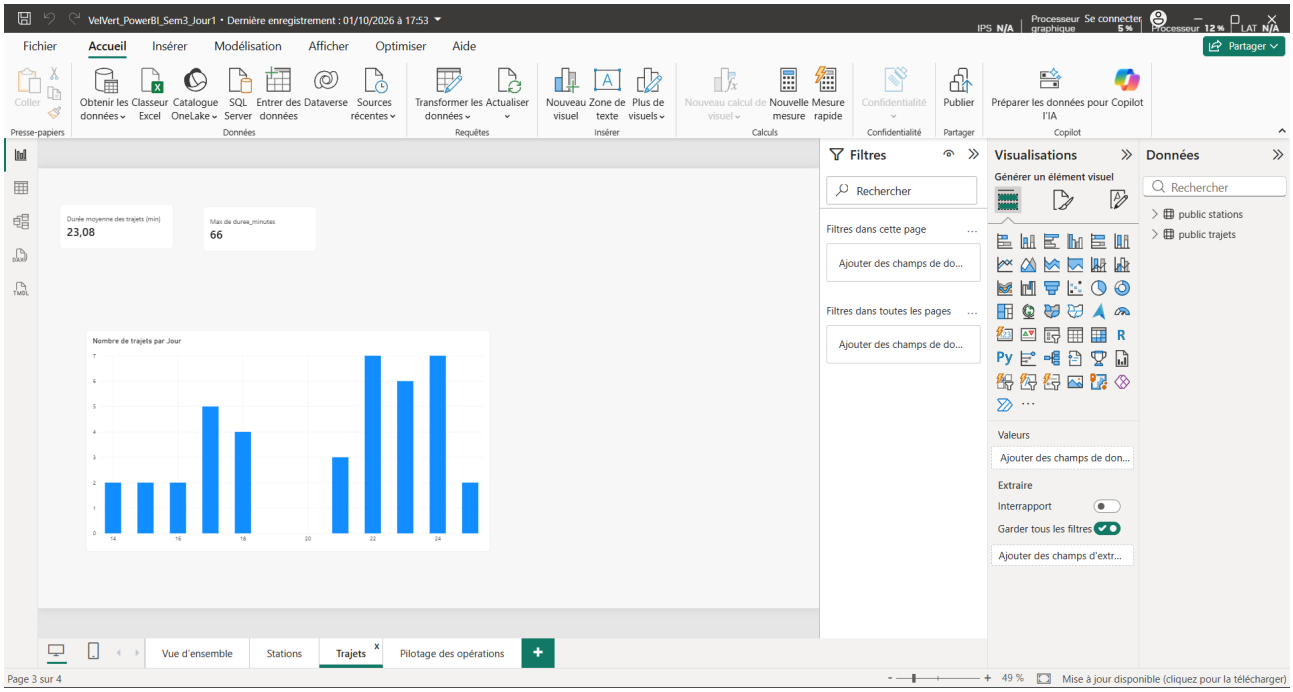


The stations page compares availability and fill rates, classifies stations by category and lists stations below the 20% availability threshold.

Decision rule

A station is considered under pressure when available bikes are less than 20% of its total capacity. This explicit rule makes the prioritisation transparent and repeatable.

5. Dashboard — Trips

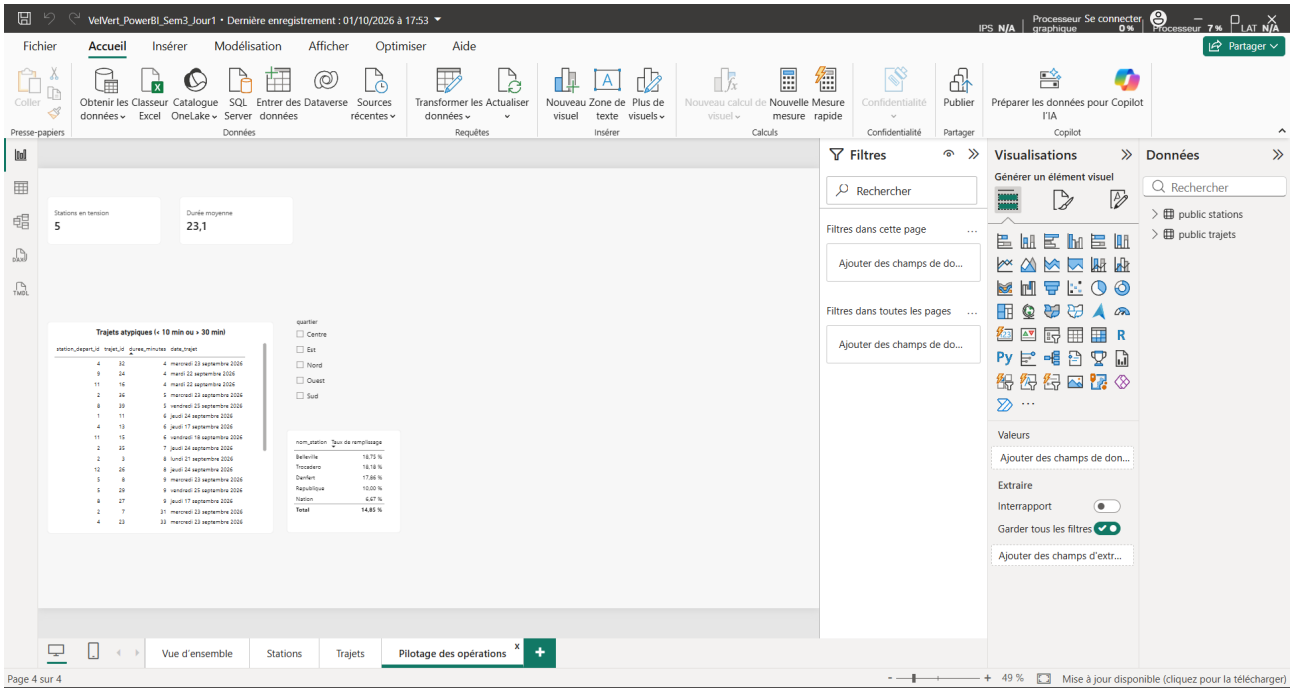


The trips page shows average and maximum duration and the daily distribution of trips. These visuals help identify patterns that may deserve further investigation.

Review threshold

The SQL exercise defines potential anomalies as trips under 10 minutes or over 40 minutes. The dashboard's atypical-trip table uses its displayed threshold; the two views should be interpreted according to their respective definitions.

6. Dashboard — Operations management



The operations page combines a table of potentially atypical trips, a neighbourhood filter and a station fill-rate summary for operational review.

Operational use

The neighbourhood filter helps the team focus its review geographically. The displayed results support triage, but do not replace local operational knowledge.

7. Recommendations and limitations

Recommended action

Start Monday’s replenishment round with the stations whose availability rate is below 20%. Order the list from the lowest availability rate to the highest. Include the station name, neighbourhood, available-bike count, capacity and calculated rate so the operations team can act on the result.

Why this criterion?

The availability rate makes stations of different capacities more comparable than a raw bike count alone. It also directly matches the operational question: which stations are most in need of replenishment?

Limitations

• The dataset is small and represents the snapshot supplied for the training exercise.

• The data do not explain why a station has few bikes or predict demand over the next hours or days.

• A very short or long trip is a potential anomaly, not proof of a data error or misuse.

• The dashboard reflects the data loaded for the exercise; it is not a live monitoring service.

Conclusion

This project demonstrates a practical workflow from business question to SQL query, result checking and dashboard-based communication. It shows how a simple, documented criterion can turn station data into a clear operational priority.

Project type	Tools	Training
Operational data analysis	**SQL · PostgreSQL · Power BI · Python**	**Simplon Data Analyst Programme, 2026**